

**ZCAMS**

Zambia Customs Agent Management System

ZCAMS ASYCUDA API Specification

Proposed read-only integration endpoints for sharing Z-SAD, customs, company, invoice, and payment status data.

Document	ZCAMS ASYCUDA API Specification
Audience	ASYCUDA integration teams, ZAFFA platform owners, and ZCAMS engineers
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Theme	ZCAMS green, gold, orange, and light operational panels

This document uses designed visual panels instead of live screenshots. The panels are intentionally high-level so they can be shared with users, ZAFFA stakeholders, and integration partners without exposing live operational data.

1. Integration Overview

This document proposes eight read-only API endpoints under **/api/asycuda/v1**. The goal is to allow ASYCUDA to retrieve the Z-SAD number, customs details parsed by the CFA, company profile information, generated invoice details, and whether the invoice is paid.

1. ZCAMS BL	2. Z-SAD	3. Invoice	4. Payment	5. ASYCUDA Query
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Security requirement	Recommendation
Transport	HTTPS only. Reject plain HTTP in production.
Authentication	Server-to-server API key or signed bearer token issued by ZAFFA/ZCAMS.
Authorization	Read-only scope: asycuda:read. No user password sharing.
Audit	Log request id, endpoint, caller id, timestamp, query, and response status.
Data ownership	ZCAMS remains source of operational CFA workflow; ASYCUDA consumes agreed declaration/payment data.

2. Proposed Endpoint Catalogue

#	Method	Endpoint	Purpose
1	GET	/api/asycuda/v1/health	Confirm API availability and version compatibility.
2	GET	/api/asycuda/v1/declarations	Search ZCAMS declarations by Z-SAD, BL number, invoice number, company TPIN, or payment status.
3	GET	/api/asycuda/v1/declarations/{z_sad_number}	Return a complete declaration summary for one active or historical Z-SAD.
4	GET	/api/asycuda/v1/declarations/{z_sad_number}/company-profile	Share CFA company registration and compliance profile details linked to the declaration.
5	GET	/api/asycuda/v1/declarations/{z_sad_number}/customs-details	Share parsed customs and shipment fields from the Bill of Lading.
6	GET	/api/asycuda/v1/declarations/{z_sad_number}/cargo	Share cargo line items, HS codes, weights, quantities, GN 83 category, and containers.
7	GET	/api/asycuda/v1/declarations/{z_sad_number}/invoice	Share the generated invoice, amount breakdown, CapitalPay reference, and PDF metadata.
8	GET	/api/asycuda/v1/declarations/{z_sad_number}/payment-status	Tell ASYCUDA whether the linked invoice has been paid and whether cargo can be released.

3. Endpoint Details And Examples

1. GET /api/asycuda/v1/health

Purpose: Confirm API availability and version compatibility.

Query parameters: None

Success response example:

```
{
  "status": "ok",
  "system": "ZCAMS",
  "version": "v1",
  "timestamp": "2026-05-29T09:00:00Z"
}
```

2. GET /api/asycuda/v1/declarations

Purpose: Search ZCAMS declarations by Z-SAD, BL number, invoice number, company TPIN, or payment status.

Query parameters: z_sad, bl_number, invoice_number, company_tpin, status, payment_status, page, page_size

Success response example:

```
{
  "items": [
    {
      "z_sad_number": "Z-SAD-3322-123456ABCDEF",
      "bl_number": "MSC4051473322",
      "invoice_status": "AWAITING_PAYMENT"
    }
  ],
  "page": 1,
  "page_size": 25
}
```

3. GET /api/asycuda/v1/declarations/{z_sad_number}

Purpose: Return a complete declaration summary for one active or historical Z-SAD.

Query parameters: Path parameter only

Success response example:

```
{
  "z_sad_number": "Z-SAD-3322-123456ABCDEF",
  "reviewed_status": "AWAITING_PAYMENT",
  "company_id": "company-zaffa-demo",
  "invoice_number": "INV-20260529120000-A1B2C3"
}
```

4. GET /api/asycuda/v1/declarations/{z_sad_number}/company-profile

Purpose: Share CFA company registration and compliance profile details linked to the declaration.

Query parameters: Path parameter only

Success response example:

```
{
  "company": {
    "name": "ZAFFA Clearing & Forwarding",
    "tpin": "1000123456",
    "zra_licence": "ZRA-CFA-2026",
    "zaffa_number": "ZAFFA-001"
  }
}
```

5. GET /api/asycuda/v1/declarations/{z_sad_number}/customs-details

Purpose: Share parsed customs and shipment fields from the Bill of Lading.

Query parameters: Path parameter only

Success response example:

```
{
  "bl_number": "MSC4051473322",
  "consignee_tin": "1000123456",
  "route_type": "IMPORT",
  "transport_mode": "SEA",
  "zra_regime": "IMPORT_HOME_USE"
}
```

6. GET /api/asycuda/v1/declarations/{z_sad_number}/cargo

Purpose: Share cargo line items, HS codes, weights, quantities, GN 83 category, and containers.

Query parameters: Path parameter only

Success response example:

```
{
  "cargo_items": [
    {
      "description": "Motor vehicles",
      "hs_code": "8703",
      "weight": 12000,
      "gn83_category": "MOTOR_VEHICLE"
    }
  ],
  "containers": [
    {
      "container_no": "MSCU1234567",
      "size": "40HC"
    }
  ]
}
```

7. GET /api/asycuda/v1/declarations/{z_sad_number}/invoice

Purpose: Share the generated invoice, amount breakdown, CapitalPay reference, and PDF metadata.

Query parameters: Path parameter only

Success response example:

```
{
  "invoice_number": "INV-20260529120000-A1B2C3",
  "invoice_type": "SERVICE_FEE_ONLY",
  "total": 35.0,
  "currency": "USD",
  "pdf_url": "https://zcams.info/download/invoice/inv-id.pdf"
}
```

8. GET /api/asycuda/v1/declarations/{z_sad_number}/payment-status

Purpose: Tell ASYCUDA whether the linked invoice has been paid and whether cargo can be released.

Query parameters: Path parameter only

Success response example:

```
{
  "invoice_status": "SETTLED",
  "payment_status": "SETTLED",
  "settled_at": "2026-05-29T09:15:00Z",
  "release_status": "CARGO_RELEASED"
}
```

4. Common Response And Error Model

HTTP status	Meaning	Example use
200	Success	Declaration, invoice, cargo, or payment status returned.
400	Bad request	Invalid date, status, or malformed Z-SAD number.
401	Unauthenticated	Missing or invalid ASYCUDA API key/token.
403	Forbidden	Caller does not have asycuda:read scope.
404	Not found	Z-SAD, invoice, or company profile not found.
409	Conflict	Z-SAD exists but is retired/superseded and active-only data was requested.
429	Rate limited	Too many requests from the integration client.
500	Server error	Unexpected ZCAMS failure with request id for support tracing.

Standard error body:

```
{
  "error": {
    "code": "DECLARATION_NOT_FOUND",
    "message": "No ZCAMS declaration was found for the supplied Z-SAD number.",
    "request_id": "req_20260529_001"
  }
}
```

5. Future Write-Back Option

If ASYCUDA later needs to send processing acknowledgements into ZCAMS, add a separate authenticated write-back endpoint such as **POST /api/asycuda/v1/declarations/{z_sad_number}/acknowledgement**. That endpoint should persist ASYCUDA reference numbers, filing status, timestamps, and rejection reasons in a dedicated audit table before any UI status is changed.